Transmission Rate-based Congestion Control in Vehicular Ad Hoc Networks

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Abstract: Innovations in technology and its rapid deployment in day-to-day matters has become the center of attraction for improvingquality of life. In this context, Vehicular Ad Hoc Network(VANET) has become a promising topic in research communitydue to its various applications like safety, infotainment, traffic management application etc. and supports for Vehicle-to-Vehicle(V2V) and Vehicle-to-Infrastructure (V2I) communication. To reducethe number of accidents, safety applications require effectiveV2V communication. Vehicles broadcast two types of messages,beacons and emergency messages, through the Control channel (CHH) of Dedicated Short Range Communication (DSRC).Beacon messages include vehicle status like position, velocity, etc.that are disseminated periodically. With the increase in a numberof vehicles, dissemination of safety messages becomes congested,for mitigation of which we analyze congestion detection schemes as well as congestion control. Effective congestion control algorithminvolves incorporation of Decentralized Congestion Control (DCC) scheme using rate adaption on the lower layers of theprotocol stack, i.e., Physical and Medium Access layer (MAC).The effects of congestion on vehicular safety can be controlled by designing a DCC algorithm including with priority model andtransmission rate of the beacon, which provides more reliable and timely reception of safety messages. DCC algorithm controlsthe congestion by maximizing the transmission rate of the high priority safety messages. A novel approach is proposed regardingcongestion control and performance is analyzed with respect to some indexes such as Packet-Delivery-Rate (PDR) and End-to-End (E2E) delay.The proposed scheme improves the PDR and E2E dealy along varying traffic density.

Keywords: VANET, Congestion control, Congestion detection, Transmission rate

I. INTRODUCTION

Road traffic safety has been a worldwide matter due to escalation in death rate. A report from the World Health Organization (WHO) states that approximately one and half a million people died due to road accidents in 2013. Road death ratio canbe reduced by providing the traffic status, early warning, etc.to the vehicle owners through safety messages. VANET[1]is growing as an enormous technologyin a wireless network that making communication possible between vehicles [2][3]. VANET consists of vehicles, GPS, road side units, radio transceivers and internet connection. It attracts lots of researchers and industries towards itself due to its application in road safety, traffic control management and entertainment. United State Federal Communications Commission (FCC) has allocated 75MHz of bandwidth spectrum in 5.9 GHz band. Vehicles in the network can transfer data at a rate of 6-27 Mbps and the range of communication is up to 1000m[4][5].Safety applications envisaged to support life-critical safety message dissemination such as intersection collision, emergency electronic brake lights, safety warning, lane change assistant etc. For safety application, the foremost requirements are strict delay sensitive and reliability. Decision making has to be done in few milliseconds and therefore in other words you can say it has hard deadline for message delivery. Medium Access layer (MAC) protocols play a major role in scheduling safety messages transmissions, in-order to deliver the message in a short time [6][7]. At MAC layer vehicle transmit two types of messages that are periodic (beacons) and event driven messages. As the density of vehicle increases, the number of beacon transmission also increases results congestion in the communication channel.Congestion can have a significant impact on the effectiveness of the safety applications, as it increase latency and decrease reliability for highly critical message (event driven) transmission. To mitigate this issue, it is very important to detect and control the congestion. In this paper, to control the congestion, the beacons are prioritized in the queue and transmit with high transmission rate. LIMERIC algorithm is used to increase the rate of transmission.

The rest of the paper is organized as follows. Section 2 discussed the background including motivation, MAC layer protocols, congestion detection and control. Section 3 described the problem formulation and proposed algorithm. Section 4 shows the simulation results. Finally, section 5 concludes the paper.

II. BACKGROUND

A. Motivation

Now a day's continuously increasing traffic has led to several problems such as road accidents, traffic jams and time delay,

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etc. However, passenger safety is a critical one because the ratio of road accidents is increasing day by day. In 2010, National Crime Records Bureau reported that deaths due to road accidents are 42.8% of total deaths. To resolve this issue, automobile companies and government have adopted many measures such as Intelligent Transportation System (ITS), traffic management, and many other safety applications. VANET came as a key part of ITS, which rises to smart vehicles. The communication between vehicles may rise to the problem of packet congestion and loss of data. Message congestion generally occurs at MAC layer. There are lots of MAC layer protocols based on contention-free and contention based techniques. Many flaws exist in these protocols, and there is always a need to achieve the maximum efficiency incommunication. The dynamic nature of vehicles poses many challenges of congestion in communication. So it is required to design a congestion control protocols for VANET.

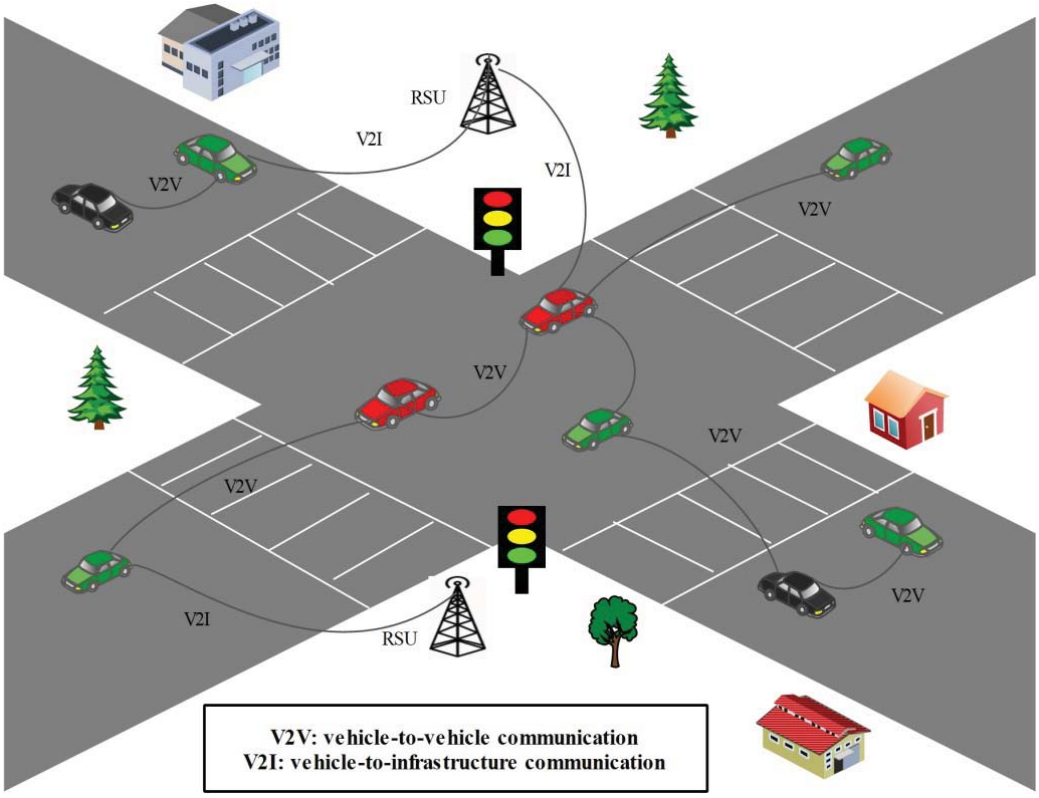


Fig. 1. VANET architecture

B. MAC layer

MAC protocol is needed for efficient utilization of medium through fair allocation of channel. To enhance the V2V communication, vehicle should choose a best time slot for data transmission. MAC protocols are classified accordingto the channel access method for V2V communication. The requirement of safety messages is reliability and lowlatency. There are two types of safety messages disseminated by each vehicle at MAC layer that are Periodic (Beacons)[8] and Event-driven messages.Beacons are the messages disseminated by the vehicle periodically about VANETstatus information like speed, position, velocity, etc. Event-driven messages are disseminated whenever some critical position occurs like accident (collision ahead), blind spot, lane change, etc. As the density of vehicle increases the number of beacons also increases that result to congestion in the communication channel. Sometimes, due to congestion the highly critical messages get lost or reach their destination with delay. Therefore, it is necessary to control the congestion for the effectiveness safety applications. A key challenge related to this beaconing activity is to control the channel load to avoid channel congestion and better reception of emergency messages.

C. Congestion Detection

Under the situation of congestion, it is required to detect the congestion first. After identifying the congestion location, broadcast the congestion information to vehicle drivers those chasing towards them. The phases of congestion detection are described as follows.

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Information gathering - The most basic phase of congestion detection is to gather vehicle status of the vehicle's in the communication range. OBU are equipped on each vehicle, which helps in gathering information such as location, velocity, etc.

- Information sharing – In this phase the congestion detection scheme will decide, which to be broadcast according to the importance of data and channel bandwidth.
- Information consolidated - During this phase, the information collected from the vehicles in the communication range and remove inconsistent information.
- Knowledge creation - This is the final phase, where the information is converted into knowledge. This knowledge is relayed to vehicle's operator showing the congestion area.
- Knowledge Propagation - This is essential because it helps other vehicle reaching towards the affected area. However, knowledge is more useful than raw information and it does not overload the bandwidth.

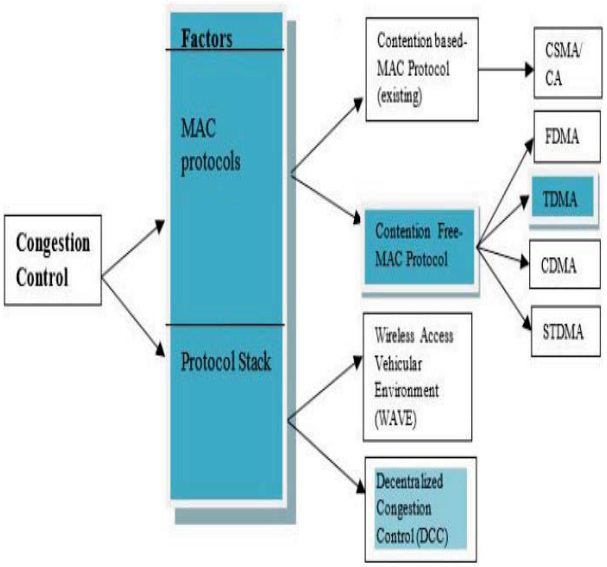


Fig. 2: Congestion Control Schemes

D. Congestion Control

Congestion control in the communication networkguarantees high reliability and undelay delivery of safety messages[9]. Congestion is controlled in twoways i.e. by replacing the existing standard MAC protocol (CSMA/CA) by Time Division Multiple Access protocol (TDMA)and using Decentralized Congestion Control (DCC)[10]as shown in Fig.2. However, TDMA MAC protocol[11] also needs a centralized authority to allocate and manage time-slots. In terms of reliable communication, centralized topology is not much suitable for VANET.Therefore, Decentralized Congestion Control (DCC) as shown inFigure 2. However, TDMA MAC protocol also needs a centralizedauthority to allocate and manage time-slots. In termsof reliable communication,centralized topology is not muchsuitable for VANET. Therefore, Decentralized CongestionControl (DCC) protocol stack is used in this paper.protocol stack is this paper.

III. PROPOSED SCHEME

In this paper, the system model comprises of N number of vehicles arriving equipped with Global Positioning System (GPS) receivers and digital maps.Assuming set of nodes N= v_1, v_2, ……,$v_n$assigned multiple applications i.e. independent to each other. Messages are queued at MAC sublayer i.e. extension of MAC layer before transmission. The proposed work emphasis is on the reliable and time delivery of emergency messages through V2V communication. Channel model adopted is Nakagami-m distribution used for fading radio wave propagation. The rate of transmission of beacon messages plays a vital role in congestion control. Asthe transmission rate increases, the total time of delivery will decreases. Therefore, the objective of the problem is to maximize the rate of transmission while keeping the network load below the maximum beaconing threshold (MBL).

$$max_{TR \in TR} \ (min_{v_{i \in N}} TR(i))$$

Subject to

$$BL(TR,i) \leq MBL$$

Where TR is transmission Rate of beacon messages and BL is beacon load. DCC queue contains beacon messages generated by the vehicles. The proposed scheme considers of two types of DCC queue: queue safety and queue othermsg. The first queue carries safety messages and the second contains other miscellaneous messages. We consider queue_safety only and discard the queue_othermsg. First schedule the beacon messages, considering the adaptation of priorities and transmissionrate. The messages will be dequeued by following priority queue model. Whenever the message is dequeued, apply LIMERIC [12]algorithm to increase the transmission rate of the message. In this way, the beacon having the high priority transmits first with a significant transmission time without occupying the channel for a long time.

IV. SIMULATION RESULTS

In this section, we compute the performance of the proposed algorithm with the help of the network simulator (NS2.35) and mobility simulator (SUMO). It is assumed that all the vehicles are equipped with GPS and wireless communication interface DSRC. Each vehicle generates ten beacons per second andis of fixed size (512bytes).

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Algorithm 1 : The proposed algorithm

Input: queue safety, queue othermsg
Output: Message dissemination with high transmission rate
1: Upon receiving messages scheduler groups all the messages in queue (queue safety, queue othermsg) of same type according to the priority.
2: queue safety _ GetmsgNodelocal (M,N)
3: queue othermsg _ GetmsgNodelocal(M,N)
4: Select queue safety and discard queue othermsg
5: Set the priority of Beacon messages
6: while queue safety 6= NULL do
7: high priority msg = Max (priority queue safety, N)
8: Apply LIMERIC scheme on high proritymsg
9: end while
10: Maximize the transmission rate of high priority messagein order to minimize the congestion

The scenario taken in proposed algorithm is bi-directional road with 5-60 veh/km/lane. Vehicle move at the speed of 10 Km/h to 90 Km/h. The network parameters used in the simulation is described in Table I.

TABLE I: Network Parameters

Parameters	Values
Area	4000 m _ 4000 m
Simulation time	500 seconds
Number of vehicles	50 - 250
Number of lanes	2
Velocity of vehicles	10 kmph □ 90 kmph
Transmission range	400 m
Size of packets	512 bytes
802.11p Data rate	3 Mbps
Radio Propagation model	Nakagami-m
Mobility model	freeway

The performance of the proposed scheme us evaluated based on PDR and E2E delay.
PDR: The ratio of number of successfully delivered messages by total number of messages sent by the source vehicle.
E2E delay: Total time taken by the beacon message to get received by the receiver.

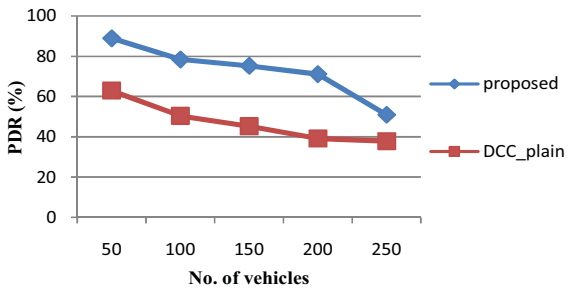


Fig. 3. PDR versus density of vehicles

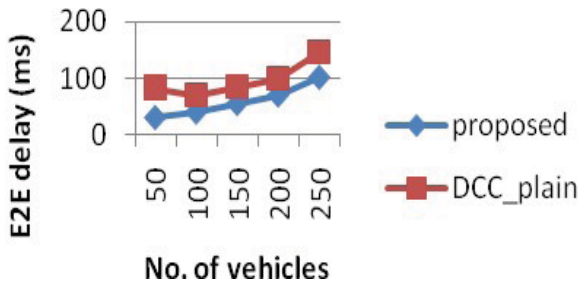


Fig. 4. E2E delay versus density of vehicles

As shown in Figure 3, the PDR decreases as the vehicle density increases. In low density of vehicle, the PDR is high due to less number of beacon messages. As the vehicle density increases, the transmission of beacon increases simultaneously that results congestion in the communication channel. PDR of proposed scheme is higher than Plain-DCC because of the priority model used for the selection of messages and maximizing the transmission rate of the safety messages.

In Figure 4 the delay increases as the number of vehicles increases. E2E delay of the proposed scheme is lower than the DCC plain scheme because message transmit with fast rate. Therefore in the proposed scheme, message reaches at the receiver earlier.

V. CONCLUSION

With the increase in the number of road fatalities around the world, there is a high need for reliable congestion control scheme for efficient V2V communication. DCC is a prominent approach of congestion control at the MAC layer. It is a challenging task to design a reliable and undelayed DCC scheme under high vehicle density. The proposed scheme with priority model and maximizing transmission rate improves congestion control. The results show the improvement in PDR and E2ED by the proposed scheme as compared to DCC-plain.

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